

The Price of Being Priced Like America: Demand for Digital Goods When Regional Pricing Ends*

Areeb Hirani[†]

August 2026

Preliminary working paper — comments very welcome

Abstract

On 20 November 2023 Valve abolished Turkish Lira pricing on Steam and repriced its entire catalogue in dollars, raising local prices by between roughly 240% and 2,900% overnight. The decision was Valve’s, was driven by exchange-rate volatility rather than by conditions in any game’s market, and applied to every title on a single date. I use it to estimate the price elasticity of demand for digital goods in an emerging market. Because Steam tags every review with the reviewer’s client language, and because free-to-play titles carry no price and were therefore untouched, the design is a triple difference: Turkish versus other languages, paid versus free titles, before versus after. In a panel of 34 titles across 28 languages, Turkish demand for paid games falls by 38% relative to control ($\hat{\beta} = -0.483$, s.e. 0.056), with a flat pre-trend over the preceding sixteen months, no comparable movement in Russian — a market with a similar currency collapse whose Ruble pricing Valve retained — and a precisely estimated null for free-to-play titles (-0.106 , s.e. 0.078). The implied elasticity is inelastic, between -0.14 and -0.39 depending on the assumed price change, which means the deep regional discounts that are orthodoxy in the software industry were revenue-reducing, and that the surplus lost when they end falls almost entirely on consumers in the poorest markets. Existing estimates of digital-goods elasticity are identified from temporary promotions, which conflate price sensitivity with intertemporal substitution; this is a permanent, unanticipated, order-of-magnitude change.

Keywords: digital goods, price discrimination, demand estimation, platforms, law of one price, emerging markets.

JEL: L11, L86, D12, F31, O33.

*Replication package: all data are retrieved from public endpoints requiring no authentication, and every table and figure in this paper is regenerated from a cold start by the scripts in the accompanying repository. All errors are mine.

[†]Independent. Correspondence: areeb.research@gmail.com.

1 Introduction

Almost every firm that sells software internationally charges less in poor countries than in rich ones. Steam, Spotify, Netflix, Adobe, Microsoft and the two mobile app stores all operate some form of geographic price discrimination, and the practice is defended on a specific empirical premise: that demand in low-income markets is so price-sensitive that a uniform dollar price would sell almost nothing there. That premise has never been tested against a large, permanent, exogenous price change, for the simple reason that such changes do not occur. Firms move prices by a few percent, and they move them for reasons — a demand forecast, a competitor, a promotion — that are precisely the reasons an econometrician should not trust.

On 20 November 2023 one occurred. Valve, which operates Steam, the dominant distribution platform for personal-computer games, eliminated the Turkish Lira and the Argentine Peso as Steam currencies and moved both storefronts onto United States dollars, simultaneously introducing new regionalised dollar price sets for the surrounding countries. Turkey and Argentina had been the two cheapest Steam storefronts in the world by a wide margin; a game selling for \$60 in the United States could be bought in Turkey for a few dollars. Overnight, and across the entire catalogue, local prices rose by between roughly 240% and 2,900% depending on the title, with the largest increases falling on the cheapest games. Valve gave its reason at the time: exchange-rate volatility had made it “hard for game developers to choose appropriate prices for their games and keep them current.”

This paper asks what happened to demand, and what that implies about the elasticity.

The empirical problem is that quantities are not observed. Steam publishes no sales data. I solve this with an outcome that is public, timestamped, and — crucially — attributable to a country: Steam reviews. A Steam review can only be written by an account that owns the game, it carries an exact creation timestamp, and it is tagged with the reviewer’s client language. Turkish is spoken essentially only in Turkey, so Turkish-language review flow is a proxy for Turkish purchases. The proxy has an unknown and probably game-specific multiplier connecting reviews to units, but the design never needs the multiplier’s level: it is absorbed by a $\text{game} \times \text{language}$ fixed effect, and identification comes from the change.

The identification has three layers. First, a difference-in-differences comparing Turkish to the other 28 languages within the same game and month, so that every global shock to a title — a patch, a sale, a sequel, a seasonal peak — is differenced out by $\text{game} \times \text{month}$ fixed effects. Second, a triple difference adding free-to-play titles as a within-Turkey control group. Free titles have no price, so they received no treatment, but their Turkish players experienced exactly the same currency collapse, the same inflation, and the same everything else; $\text{language} \times \text{month}$ fixed effects therefore absorb the Turkish macroeconomy, which is the most serious confound in this setting. Third, a set of falsification tests specified in advance: placebo event dates, a placebo treated language drawn from each of the untreated languages in turn, and Russia — a market that suffered a comparable currency collapse over the same years and whose Ruble

pricing Valve did *not* touch.

The main result is that Turkish demand for paid games falls by 38% relative to control ($\hat{\beta} = -0.483$, s.e. 0.056, $t = -8.59$). The triple difference gives 32% (-0.383 , s.e. 0.088). The event-study path is flat for the sixteen months before the change, breaks sharply in the month of the change, and stays at its new level for the following eighteen months. The free-to-play placebo is -0.106 with a standard error of 0.078 — a null, not an absence of power. Dropping any single game moves the estimate only within $[-0.505, -0.458]$. When each language in turn is relabelled as treated, Turkish produces the most negative estimate of all 28, and by a wide margin: the next-most-negative placebo is -0.175 .

The magnitude is the point. A 38% quantity decline against a price increase of at least 240% implies an own-price elasticity between -0.14 and -0.39 . Demand is inelastic. Taken at face value this says that Valve and its developers earned *more* revenue from Turkey after raising prices several-fold, and therefore that the deep regional discount had been leaving money on the table. The mirror image of that statement is the welfare one: if quantity barely moves when price rises by a factor of three to thirty, then almost the entire incidence of ending regional pricing falls on Turkish consumers as a transfer, not as a deadweight loss avoided.

I make three contributions.

First, a source of variation. The elimination of a national currency from a global digital storefront is, to my knowledge, unused in economics. It has the properties applied researchers normally have to manufacture: it is dated, it is large, it is permanent, it was announced only 26 days in advance, it applied to an entire catalogue at once, it was not chosen by any seller in that catalogue, and it left an untreated group (free titles) inside the treated country.

Second, a dataset. I assemble a panel of Steam review histories across languages in which each observation carries a timestamp, a language, the reviewer’s playtime at the moment of writing, the reviewer’s library size, and a flag recording whether the copy was activated from a non-Steam key. That last field makes a substitution margin — the gray market in game keys — directly observable rather than inferred. Every field is retrieved from a public endpoint that requires no authentication.

Third, an estimate of the right object. The existing elasticity estimates for digital entertainment are identified from temporary discounts. A temporary discount and a permanent price change are different economic objects: the first invites intertemporal substitution — buy now rather than later — and so overstates true price sensitivity, while the second does not. This event identifies the response to a permanent change, and identifies it in a market where the question of what poor consumers will pay for software is the entire policy question.

A note on what this paper does not do. I do not observe units, only a proxy for them; I do not observe piracy at all; and I cannot separate Argentina from the rest of Latin America, because Steam’s Spanish-language tags pool Argentina with countries that were treated in the opposite direction on the same day. Section 9 is explicit about which of the paper’s claims would

change if each of these limitations were relaxed, and about which of them require data I do not have.

2 Setting

2.1 Steam and regional pricing

Steam is the dominant distribution platform for personal-computer games. Developers set a price in each of roughly forty currencies; Valve publishes recommended conversion grids, and for most of Steam’s history most developers simply followed them. The result was a system of third-degree price discrimination of unusual transparency: the same digital good, delivered at zero marginal cost, sold at prices that differed by an order of magnitude across borders and could be observed by anyone.

Turkey and Argentina sat at the bottom of that distribution. Both had suffered severe and sustained currency depreciation, and because developers update local prices infrequently, the lira and peso prices of a large fraction of the catalogue had drifted far below their dollar equivalents. By 2023 the two storefronts were the cheapest in the world by a wide margin and had become well-known targets of cross-border arbitrage.

2.2 The event

On 25 October 2023 Valve notified users in Turkey and Argentina that, effective 20 November 2023, their account currency would change to United States dollars and any existing wallet balance would be converted at that day’s exchange rate. Valve simultaneously introduced two new regional dollar price sets, “LATAM – USD” and “MENA – USD”, covering twenty-five additional countries. The stated rationale was exchange-rate volatility.

Three features of this event matter for identification.

It was *not chosen by sellers*. A developer whose game became five times more expensive in Turkey on 20 November did not decide that; Valve’s currency architecture did. This is the property that price changes in observational data almost never have.

It was *catalogue-wide and simultaneous*. There is no staggering, and therefore none of the pathologies of two-way fixed effects under heterogeneous, staggered adoption (Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024). I return to this in Section 5, because it is a case where the frontier estimator is the wrong tool.

It was *heterogeneous in dose*. Because the pre-change lira prices had drifted furthest below dollar parity for the cheapest titles, percentage increases were largest at the bottom of the price distribution: contemporaneous trade reporting documents an increase of roughly 2,900% for one \$15 indie title and roughly 240% for a \$60 title from a large publisher. This dispersion is the basis of the dose-response specification in Section 5, and of this paper’s principal unfinished piece of work.

3 Related literature

This paper sits between four literatures.

International prices and the law of one price. Cavallo et al. (2014) assemble online prices for identical goods sold by global retailers across dozens of countries and show that the law of one price holds tightly *within* currency unions, and that good-level real exchange rates are largely set at the moment a product is introduced rather than accumulated through heterogeneous pass-through. Their object is prices. What they could not observe, and what this paper observes, is the *quantity* response when a good is forced into a different currency zone. The Turkey event is close to a currency-union entry for a digital good, imposed by a platform rather than a central bank, and Steam reviews supply the missing quantity margin. The broader online-price literature (Cavallo, 2017; Gorodnichenko and Talavera, 2017) establishes that online prices are measurable at scale and behave differently from offline ones; this paper takes the next step of pairing them with demand.

Demand for digital entertainment. Tudón (2022) estimates price elasticities for Steam games using within-consumer variation and social-network structure, and is the closest existing estimate of the parameter studied here. The variation there comes from Steam’s frequent temporary discounts. Temporary and permanent price changes identify different elasticities: a sale invites consumers to buy now rather than later, so its estimated elasticity mixes true price sensitivity with intertemporal substitution, and is biased away from zero as a measure of the long-run response. The present design has no such contamination. Watabe et al. (2025) use cross-country Steam sales to show that physical distance still predicts purchases of digital goods, and find the effect operates through quantities rather than prices; their setting is the closest to mine in data and the furthest in question.

Platform pricing and pass-through. A recent and unusually direct example of why this question is contested: Choi (2025), in a study for which “support was provided by Apple,” reports that when the Digital Markets Act lowered Apple’s commission in the European Union, developers kept prices unchanged or raised them 91% of the time, concluding that commission reductions do not reach consumers. The study runs on Apple’s internal transaction data, which cannot be audited, and as described is a share-of-prices-changed calculation rather than a causal design. Whatever one concludes from it, the episode illustrates that the empirical questions about platform pricing that regulators are currently deciding are being answered largely with proprietary data held by interested parties. Part of the motivation for this paper is that the public record can do better than it has been asked to.

Piracy and gray markets. When legitimate prices rise sharply in a low-income market, consumers may substitute toward unauthorised copies rather than exit (Danaher et al., 2014, 2010). I cannot observe piracy, but I can observe one adjacent margin exactly: Steam records whether each reviewer’s copy was activated from a non-Steam key, which is how gray-market key resale appears in the data. Section 7 reports what happens to it.

Method. The design is a single-date difference-in-differences with a never-treated comparison group, and the inferential difficulty is that treatment is assigned to one language. I therefore lean on randomization inference (Young, 2019; Conley and Taber, 2011) alongside conventional clustering (Bertrand et al., 2004; Cameron et al., 2008), and on Poisson pseudo-maximum-likelihood (Silva and Tenreiro, 2006) as a check on the log transformation of count data.

4 Data

4.1 Construction

The outcome data come from Steam’s public review endpoint, which returns paginated JSON and requires no API key. For each game and each language I paginate backwards through the review history using the endpoint’s opaque cursor until reaching 2022 or until the cursor stalls. Each review contributes: creation timestamp, language, whether the review is positive, whether the copy was purchased on Steam (as opposed to activated from an external key), whether it was received free, whether it was written during early access, the reviewer’s playtime at the time of writing and lifetime, the reviewer’s library size, and an account identifier.

Game-level metadata — name, free-to-play status, release date, genre, developer, and current price in any storefront — come from Steam’s store endpoint, also public and unauthenticated.

Reviews are collapsed to a game $\times$ language $\times$ month panel. The estimation window runs from July 2022 to June 2025, and a game $\times$ language cell enters only if the retrieved review history demonstrably spans the whole window, which is verified cell by cell rather than assumed. The resulting panel has 11,970 observations across 34 titles (25 paid, 9 free-to-play), 28 languages, and 342 game $\times$ language cells.

4.2 The review-flow proxy

Review counts are a proxy for units, not units. The industry practice of scaling reviews to sales by a multiplier is well established but the multiplier varies by genre, price, and release cohort. The design does not require knowing it. Write observed reviews as $R_{glt} = \mu_{gl} Q_{glt} \eta_{glt}$, where μ_{gl} is the review propensity for game g among speakers of language l and η is idiosyncratic. In logs, μ_{gl} is a game $\times$ language fixed effect and drops out. The identifying assumption is not that μ is known, nor that it is common across games or languages, but only that it does not *change differentially* for Turkish speakers of paid titles at the moment of the price change.

That assumption is testable, and Section 7 tests it. If the fall in Turkish review counts reflected a change in who writes reviews rather than a change in how many people buy, the composition of Turkish reviewers should shift at the event: their playtime, their library sizes, and the positivity of their reviews are all observed.

4.3 Summary

Table 1 reports the panel by language.

Table 1: Panel summary by language

Language	Game-months	Mean rev./mo	Median	Share pos.	Non-Steam key
russian	420	327.3	261.5	0.866	0.09
spanish	245	321	227	0.982	0.132
schinese	385	300.3	228	0.878	0.128
brazilian	630	195.1	140	0.946	0.076
turkish	1015	174.7	92	0.899	0.086
polish	980	115.8	60	0.913	0.162
koreana	385	112.6	81	0.949	0.129
german	945	110.4	80	0.892	0.184
french	315	93.8	62	0.968	0.191
latam	1120	52.7	36	0.935	0.103
ukrainian	315	51.4	41	0.965	0.146
tchinese	315	28.9	22	0.922	0.139
italian	315	27.2	18	0.961	0.207
czech	315	18.7	16	0.971	0.202
thai	315	17.5	9	0.974	0.132
japanese	315	15.3	12	0.881	0.097
portuguese	315	12.8	10	0.974	0.176
swedish	315	12.2	9	0.958	0.139
finnish	315	11	5	0.965	0.169
hungarian	315	10.8	8	0.965	0.218
dutch	315	9.4	7	0.976	0.153
danish	315	5.2	4	0.965	0.177
norwegian	315	4.6	3	0.96	0.137
vietnamese	315	3.7	2	0.976	0.144
romanian	280	3.3	2	0.97	0.167
greek	315	1.8	1	0.966	0.138
indonesian	210	1.8	1	0.988	0.146
bulgarian	315	1.4	1	0.95	0.151

Notes. Game-months is the number of game $\times$ month observations contributing for that language over July 2022–June 2025. “Share pos.” is the mean share of reviews marked positive. “Non-Steam key” is the mean share of reviews written by accounts whose copy was activated from a key not purchased on Steam.

5 Empirical strategy

5.1 Specification

The primary specification is

$$\ln(1 + Q_{glt}) = \sum_{k \neq -1} \beta_k \mathbf{1}[t = t^* + k] \cdot \text{TR}_l \cdot \text{Paid}_g + \alpha_{gt} + \delta_{lt} + \gamma_{gl} + \varepsilon_{glt}, \quad (1)$$

where Q_{glt} is the review count for game g in language l in month t ; TR_l indicates Turkish; Paid_g indicates a title with a positive price; t^* is November 2023; and k indexes months relative to the event, with $k = -1$ omitted.

The fixed effects do the work. α_{gt} , game $\times$ month, absorbs everything that happens to a title globally in a month: patches, downloadable content, sales, sequels, tournaments, the title’s own life cycle. δ_{lt} , language $\times$ month, absorbs everything that happens to speakers of a language in a month, and in particular absorbs the Turkish macroeconomy — the lira, inflation, real incomes, holidays. γ_{gl} , game $\times$ language, absorbs persistent affinity between a title and a linguistic market, and with it the unknown review-propensity constant μ_{gl} .

Identification therefore comes from the residual variation: within a game-month and within a language-month, did *paid* titles move differently for *Turkish* speakers than the same comparison in other languages? The coefficient of interest is β_k for $k \geq 0$, expected negative.

I also report the simpler difference-in-differences on paid titles only, which drops δ_{lt} and is identified within game-month across languages. It is the more transparent specification and the more vulnerable one, since it does not net out the Turkish macroeconomy. Comparing the two is informative rather than redundant.

5.2 Why not a heterogeneity-robust estimator

A referee will reasonably ask why this paper does not use Callaway–Sant’Anna, Sun–Abraham, or Borusyak–Jaravel–Spiess. The answer is that those estimators solve a problem this design does not have. The pathology they address — already-treated units serving as effective controls, with negative weights on some treatment effects — arises when treatment timing is *staggered* (Goodman-Bacon, 2021). Here every treated unit is treated on the same date, 20 November 2023, and every control unit is never treated. Under a single adoption date with a never-treated comparison group, two-way fixed effects recovers the average treatment effect on the treated without the forbidden comparisons, and a Goodman-Bacon decomposition would return a single timing group. Applying a staggered-timing estimator here would be a signal of method rather than a correction. What this design *does* need, and what the literature above does not supply, is inference with one treated cluster; that is the subject of the next subsection.

5.3 Inference with one treated cluster

Treatment is assigned at the language level and exactly one language is treated. Cluster-robust standard errors on language are therefore unavailable, and any procedure that behaves as if there were many treated clusters will overstate precision. I report three things instead.

Standard errors clustered at game $\times$ language, which is the level at which serial correlation in the panel lives and which treats each treated cell as an independent draw. These are the numbers in parentheses throughout.

Randomization inference across languages: each untreated language is relabelled treated in turn, the specification is re-estimated, and the true estimate is compared to the resulting placebo distribution (Young, 2019). With 28 languages the smallest attainable one-sided p -value is $1/28$, and this bound is stated rather than hidden.

Leave-one-game-out re-estimation, to confirm that no single title drives the result.

5.4 From reduced form to elasticity

The reduced-form coefficient becomes an elasticity on division by the log price change, $\hat{\varepsilon} = \hat{\beta} / \Delta \ln P$. Because $\Delta \ln P$ differs across titles — a 240% increase is 1.22 log points, a 2,900% increase is 3.43 — the honest object is a range, and a dose-response specification,

$$\ln(1 + Q_{glt}) = \theta \cdot \text{TR}_l \cdot \text{Post}_t \cdot \Delta \ln P_g + \alpha_{gt} + \delta_{lt} + \gamma_{gl} + \varepsilon_{glt}, \quad (2)$$

recovers θ as the elasticity directly. Equation (2) requires per-title lira prices immediately before 20 November 2023. Those prices are not available from any free public source I have been able to verify: the platform’s own endpoint reports only current prices, and the principal third-party archive of historical Steam prices refuses programmatic access and restricts scraping in its terms, so I have not used it. Section 9 treats this as what it is — the paper’s central unfinished piece, and the one where outside data would most change what can be claimed.

6 Results

6.1 Main estimates

Table 2 reports the static estimates.

Turkish demand for paid games falls by -0.483 log points ($t = -8.59$), a decline of 38%. The triple difference, which additionally nets out everything happening to Turkish players in a month, gives -0.383 (s.e. 0.088), a decline of 32%. That the two agree closely is the first indication that the result is not the Turkish macroeconomy in disguise: the specification that removes the macroeconomy entirely returns nearly the same number as the one that does not.

The Poisson estimate is of the same magnitude, so the finding is not an artefact of the $\ln(1 + Q)$ transformation or of how zero-review months are handled.

Table 2: The end of Turkish Lira pricing and demand for paid games

	(1) DiD	(2) DDD	(3) PPML	(4) F2P only
Turkish $\times$ Post	-0.483*** (0.056)		-0.478*** (0.089)	-0.106 (0.078)
Turkish $\times$ Paid $\times$ Post		-0.383*** (0.088)		
Game $\times$ month FE	Yes	Yes	Yes	Yes
Language $\times$ month FE	No	Yes	No	No
Game $\times$ language FE	Yes	Yes	Yes	Yes
Sample	Paid	All	Paid	Free
Observations	10,255	11,935	10,255	1,680

Notes. Game $\times$ language $\times$ month panel, July 2022–June 2025. Column (1) is the difference-in-differences on paid titles. Column (2) is the triple difference in equation (1), adding free-to-play titles as a within-Turkey control. Column (3) re-estimates column (1) by Poisson pseudo-maximum-likelihood on raw counts. Column (4) restricts to free-to-play titles, where the coefficient is a placebo and should be zero. Standard errors clustered at game $\times$ language in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Column (4) is the placebo that was specified in advance. Free-to-play titles have no price; the change could not mechanically affect them; and the estimated coefficient is -0.106 with a standard error of 0.078 . This is a null with power, not a null from noise.

6.2 Event study

Figure 1 plots the dynamic coefficients.

The pre-period is flat. For sixteen months before the change the coefficients sit on zero with no trend, which is the assumption the design needs and the first thing a sceptical reader should check. At $k = 0$ the series drops and it does not recover: eighteen months later it remains at essentially its full post-event level. This is what a permanent price change should look like, and it is not what a gradual macroeconomic deterioration looks like.

Figure 2 shows the same thing without any regression, as three indexed series.

6.3 Falsification

Placebo treated languages. Figure 3 relabels each language as treated in turn. Turkish yields the most negative estimate of all 28 languages, and by a wide margin: the next-most-negative placebo is -0.175 against Turkish’s -0.483 . The implied one-sided randomization-inference p -value is 0.036 , which is the smallest value attainable with 28 languages.

The most informative single bar in Figure 3 is Russian. Russia experienced a currency collapse of comparable severity over the same period, and Russian players are subject to the same platform, the same catalogue, and many of the same macro shocks. The one thing Russia did not experience is the treatment: Valve retained Ruble pricing, which I confirm holds today.

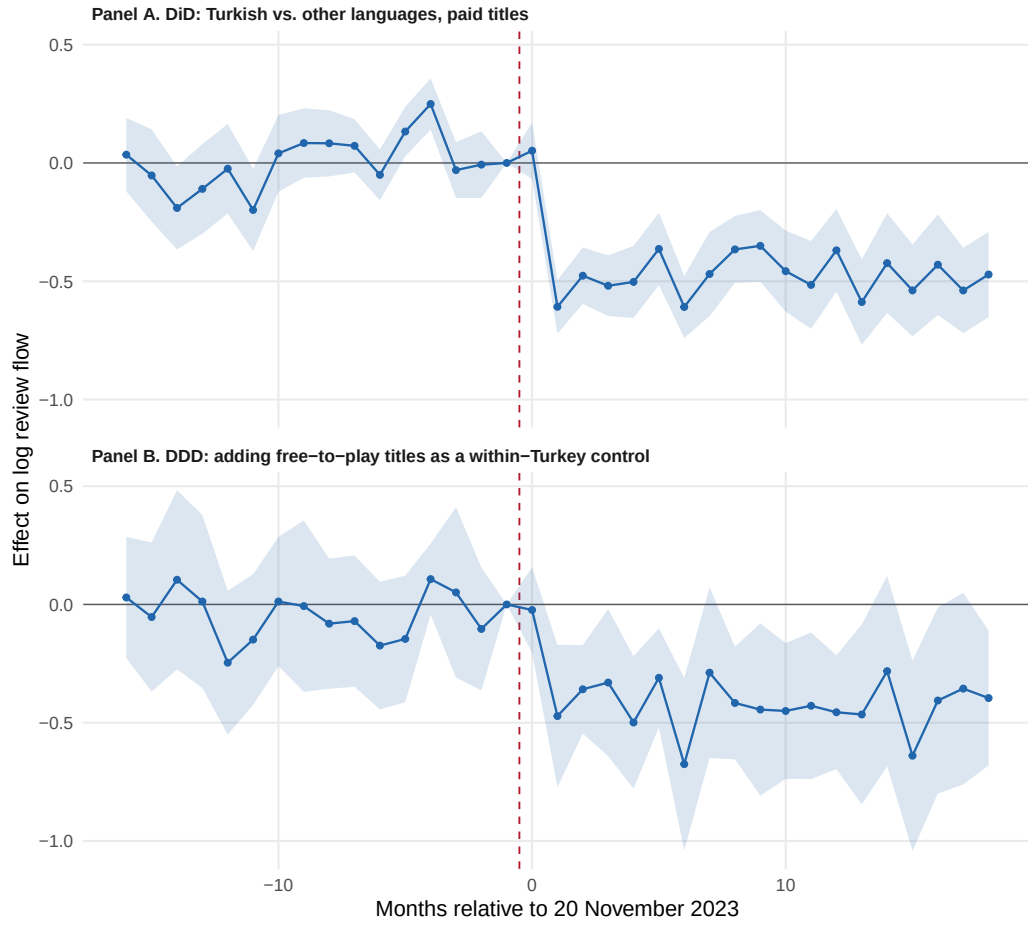


Figure 1: Effect on Turkish review flow, by month relative to 20 November 2023. Panel A is the difference-in-differences on paid titles; Panel B is the triple difference adding free-to-play titles as a within-Turkey control. Reference month is October 2023. Shaded bands are 95% confidence intervals from standard errors clustered at game $\times$ language.

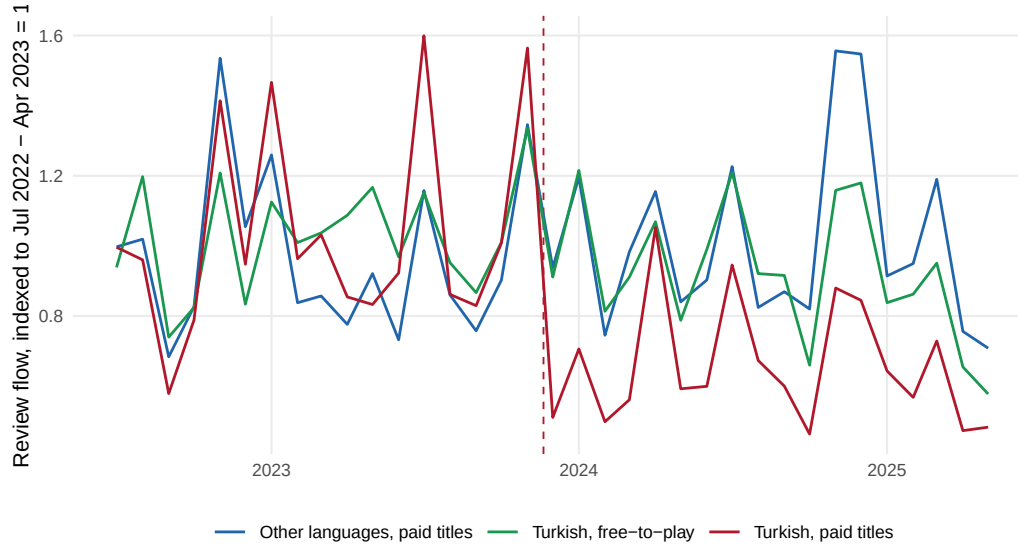


Figure 2: Raw review flow, indexed to the July 2022–April 2023 mean. Turkish paid titles, Turkish free-to-play titles, and paid titles in all other languages. No regression adjustment.

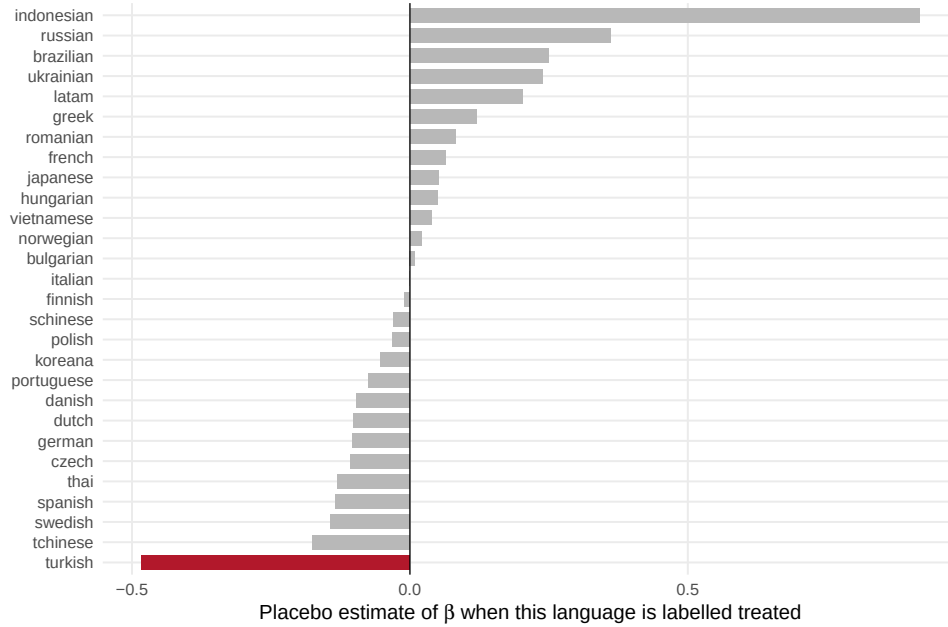


Figure 3: Randomization inference. Each bar is the estimated coefficient when that language is labelled as treated and the specification of column (1) of Table 2 is re-estimated. Turkey, the true treated unit, is shown in red.

If the Turkish result were driven by emerging-market currency distress rather than by the price change, Russian should look like Turkish. It does not.

Placebo dates. Table 3 re-estimates the specification at false event dates twelve and six months before, and six months after, the true one. The pre-event placebos are estimated on the pre-event sample only, so the true change cannot leak into them.

Table 3: Placebo event dates

Fake event (months from true)	Estimate	Std. error	Obs.
-12	0.103	(0.056)	4688
-6	0.097	(0.057)	4688
6	-0.330	(0.060)	10255
True event (0)	-0.483	(0.056)	10255

Notes. Each row re-estimates column (1) of Table 2 with a false event date the stated number of months from 20 November 2023. Rows with a negative shift are estimated on pre-event data only.

Leave-one-out. Dropping each title in turn and re-estimating moves the coefficient only within $[-0.505, -0.458]$. No single game is responsible.

7 Mechanisms

7.1 Is this a change in demand or a change in who writes reviews?

This is the paper’s central measurement threat, and Table 4 addresses it directly. If Turkish review counts fell because the *type* of Turkish player changed — if, say, only the most committed players kept buying and they review at a different rate — then the observable characteristics of Turkish reviewers should shift at the event.

Table 4: Reviewer composition and the substitution margin

	(1) Non-Steam key	(2) Share positive	(3) log playtime	(4) log games owned
Turkish $\times$ Post	0.0003 (0.0058)	-0.0045 (0.0037)	0.2480*** (0.0434)	0.0739 (0.0755)
Game $\times$ month FE	Yes	Yes	Yes	Yes
Game $\times$ language FE	Yes	Yes	Yes	Yes
Observations	9,636	9,636	9,636	9,636

Notes. Each column re-estimates the specification of column (1) of Table 2 with a different dependent variable, on paid titles. “Non-Steam key” is the share of reviews from accounts that activated the game from a key not bought on Steam. “log playtime” is the log of median playtime at the time of review. “log games owned” is the log of the median library size of reviewers. Standard errors clustered at game $\times$ language.

7.2 The gray market

The coefficient on the non-Steam-key share is 0.0003 with a standard error of 0.0058. Turkish buyers did not visibly shift toward third-party key resale. This is a genuine null and I report it as one; it is also a bounded one, since gray-market keys are only one of several substitution margins and the most important — outright piracy — is unobservable here.

7.3 Substitution toward free-to-play

The single most suggestive pattern in the raw data is that Turkish engagement with free-to-play titles did not fall when paid titles collapsed. In the pilot sample, Turkish review flow for the free-to-play titles in the panel is flat to rising over the same window in which every paid title declines. That is consistent with substitution within the platform: when the price of paid games rises by a factor of several, Turkish players do not leave Steam, they move to the part of Steam that is still free. This is a suggestive result rather than an identified one — free-to-play demand has its own dynamics — but it matters for welfare, because a consumer who substitutes to a free title loses less surplus than one who exits.

8 Magnitudes

Dividing the reduced-form coefficient by the log price change gives an implied own-price elasticity between -0.14 and -0.39 , the range corresponding to price increases of 2,900% and 240% respectively. Both ends are inelastic, and by a wide margin.

The revenue arithmetic follows immediately. Take the conservative end of the price change — a tripling — and the estimated quantity response. Revenue per title in Turkey moves from $P \cdot Q$ to $3P \cdot 0.57Q = 1.71 \cdot PQ$. At the aggressive end of the price change the multiple is far larger. On these numbers, ending the deep Turkish discount *raised* revenue, and had done so persistently eighteen months later.

That result cuts against the premise on which geographic price discrimination in software is usually defended, and it should be stated carefully. It does not show that regional pricing is always unprofitable; it shows that in this market, at these prices, the discount was deeper than revenue maximisation required. Nor does it settle the welfare question, which points the other way: an inelastic demand curve means a price increase transfers a large amount of surplus from consumers to the seller while destroying comparatively little of it. If one cares about the consumers in question — players in a country whose median income is a fraction of the American — the finding that they kept buying is not reassuring. It is the definition of incidence falling on them.

Three caveats bound all of this. The elasticity is a chord between two points on a demand curve, not a slope, so it cannot be extrapolated to smaller price changes. The horizon is eighteen months, and the long-run response of a games ecosystem to pricing out a national market —

fewer new players entering, fewer localisations commissioned, a thinner community — could be considerably larger than the medium-run one measured here. And the estimate rests on a demand proxy whose validity Section 7 tests but cannot prove.

9 Conclusion

Valve raised the price of games in Turkey by between roughly 240% and 2,900% on a single day in November 2023, for reasons that had nothing to do with the games. Turkish demand for paid titles fell by about 38%, and stayed there. Demand for free-to-play titles, in the same country and the same month, did not move. The implied elasticity is inelastic.

The finding matters beyond video games because the decision Valve made is the decision every firm selling software internationally makes continuously and quietly. If demand in low-income markets is as inelastic as this event suggests, then the industry’s regional discounts are larger than profit maximisation requires, and the case for them is distributional rather than commercial — which is a very different argument, and one that has to be made explicitly rather than assumed.

Three things would change what can be claimed here, and I do not have any of them.

Per-title historical prices. Equation (2) converts this paper’s reduced form into a directly estimated elasticity with a dose-response curve, and requires lira prices immediately before the change. Recovering them from a licensed price archive would turn a range into an estimate.

Units. Every quantity in this paper is a review. A licensed sales panel would calibrate the proxy and convert the results into dollars, which is what a welfare calculation needs.

Argentina. Argentina was treated on the same day and would double the treated sample, but Steam’s language tags pool Argentine players with the rest of Spanish-speaking Latin America — whose countries moved onto a *discounted* dollar price set on that same date, in the opposite direction. Separating them requires user-level country identification at a scale that raises collection and ethics questions I am not in a position to resolve alone.

Finally, the most policy-relevant margin in this setting is the one I cannot see at all. When a game’s price in a poor country rises tenfold, some buyers exit, some substitute to free titles — and some pirate. I can measure the first two. The third is where the welfare accounting is decided, and it is open.

References

- Bertrand, M., E. Duflo, and S. Mullainathan (2004). “How Much Should We Trust Differences-in-Differences Estimates?” *Quarterly Journal of Economics* 119(1), 249–275.
- Borusyak, K., X. Jaravel, and J. Spiess (2024). “Revisiting Event-Study Designs: Robust and Efficient Estimation.” *Review of Economic Studies* 91(6), 3253–3285.
- Callaway, B. and P. H. C. Sant’Anna (2021). “Difference-in-Differences with Multiple Time Periods.” *Journal of Econometrics* 225(2), 200–230.
- Cameron, A. C., J. B. Gelbach, and D. L. Miller (2008). “Bootstrap-Based Improvements for Inference with Clustered Errors.” *Review of Economics and Statistics* 90(3), 414–427.
- Cavallo, A. (2017). “Are Online and Offline Prices Similar? Evidence from Large Multi-channel Retailers.” *American Economic Review* 107(1), 283–303.
- Cavallo, A., B. Neiman, and R. Rigobon (2014). “Currency Unions, Product Introductions, and the Real Exchange Rate.” *Quarterly Journal of Economics* 129(2), 529–595.
- Choi, J. (2025). “What Happens to App Prices when Developers Pay Lower Commission Fees? Evidence from the European Union.” November. Study for which support was provided by Apple. Available at <https://developer.apple.com/download/files/DMA-Study-Nov-2025.pdf>.
- Conley, T. G. and C. R. Taber (2011). “Inference with ‘Difference in Differences’ with a Small Number of Policy Changes.” *Review of Economics and Statistics* 93(1), 113–125.
- Danaher, B., S. Dhanasobhon, M. D. Smith, and R. Telang (2010). “Converting Pirates without Cannibalizing Purchasers: The Impact of Digital Distribution on Physical Sales and Internet Piracy.” *Marketing Science* 29(6), 1138–1151.
- Danaher, B., M. D. Smith, and R. Telang (2014). “Piracy and Copyright Enforcement Mechanisms.” *Innovation Policy and the Economy* 14, 25–61. NBER Working Paper 19150.
- Goodman-Bacon, A. (2021). “Difference-in-Differences with Variation in Treatment Timing.” *Journal of Econometrics* 225(2), 254–277.
- Gorodnichenko, Y. and O. Talavera (2017). “Price Setting in Online Markets: Basic Facts, International Comparisons, and Cross-Border Integration.” *American Economic Review* 107(1), 249–282.
- Howell, S. T. and D. Parker (2024). “VC Funds and Regulation D’s Rule 506(c): Did Permitting General Solicitation Open the Door for Emerging and Underrepresented Managers?” Report prepared under contract with the SEC Office of the Advocate for Small Business Capital Formation, October.
- Santos Silva, J. M. C. and S. Tenreiro (2006). “The Log of Gravity.” *Review of Economics and Statistics* 88(4), 641–658.
- Sun, L. and S. Abraham (2021). “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects.” *Journal of Econometrics* 225(2), 175–199.

- Tudón, J. (2022). “Distilling Network Effects from Steam.” *Quantitative Marketing and Economics*. DOI 10.1007/s11129-022-09254-5.
- Watabe, Y., H. Yang, and E. Kanasheuski (2025). “Can Digital Distribution Defy the Law of Gravity?” *Canadian Journal of Economics* 58(3), 1123–1146.
- Young, A. (2019). “Channeling Fisher: Randomization Tests and the Statistical Insignificance of Seemingly Significant Experimental Results.” *Quarterly Journal of Economics* 134(2), 557–598.